

Model evaluation

Train, test, and the cost of a mistake

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How well does the model work?

Training data

Fit the model



Test data

Evaluate predictions

Compare predictions with reality

		Predicted	
		Positive	Negative
Actual	Positive	Correct	Mistake
	Negative	Mistake	Correct

Six stops: how to evaluate classification Model?

01

Split

Fit on one set. Judge on another.

02

Cross-validation

Every row is tested once.

03

Confusion matrix

Four kinds of answer.

04

Metrics

Accuracy is only one of them.

05

Thresholds

A probability is not yet a class.

06

ROC and AUC

Every cutoff, in one picture.

FROM LAST WEEK

We can classify. Is the classifier useful?

Logistic regression, LDA, QDA, and KNN can disagree. Today we measure the disagreement.

Three methods, one scorecard

Logistic

A probability

Then a cutoff turns it into a class.

LDA / QDA

A class shape

Then the closer shape gets the label.

KNN

A vote

Nearby training rows decide the class.

PART 1

Train and test

The model learns on one set. The grade comes from another.

Hold something out

1 Fit, then grade



Train, 70%. The model learns these rows.

Test, 30%.

Stand-ins for new customers.

2 Choose a setting, then grade



Every choice is finished before the test rows are opened.

Three settings that matter

0.25

test_size

The share held out. Often 0.2 to 0.3.

0

random_state

Fixes the shuffle, so the split can be repeated.

score

On the test set

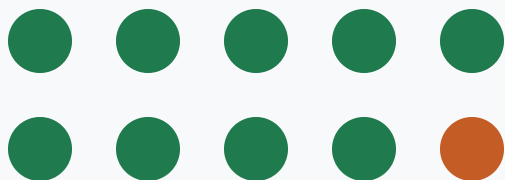
For a classifier, the share of test rows labeled correctly.

One split can look lucky

EXAMPLE, SPLIT A

0.90

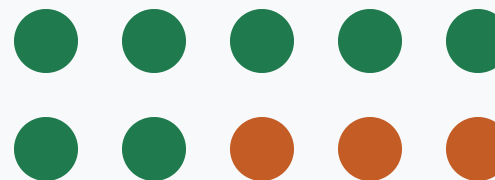
9 of 10 test rows correct



EXAMPLE, SPLIT B

0.70

7 of 10 test rows correct

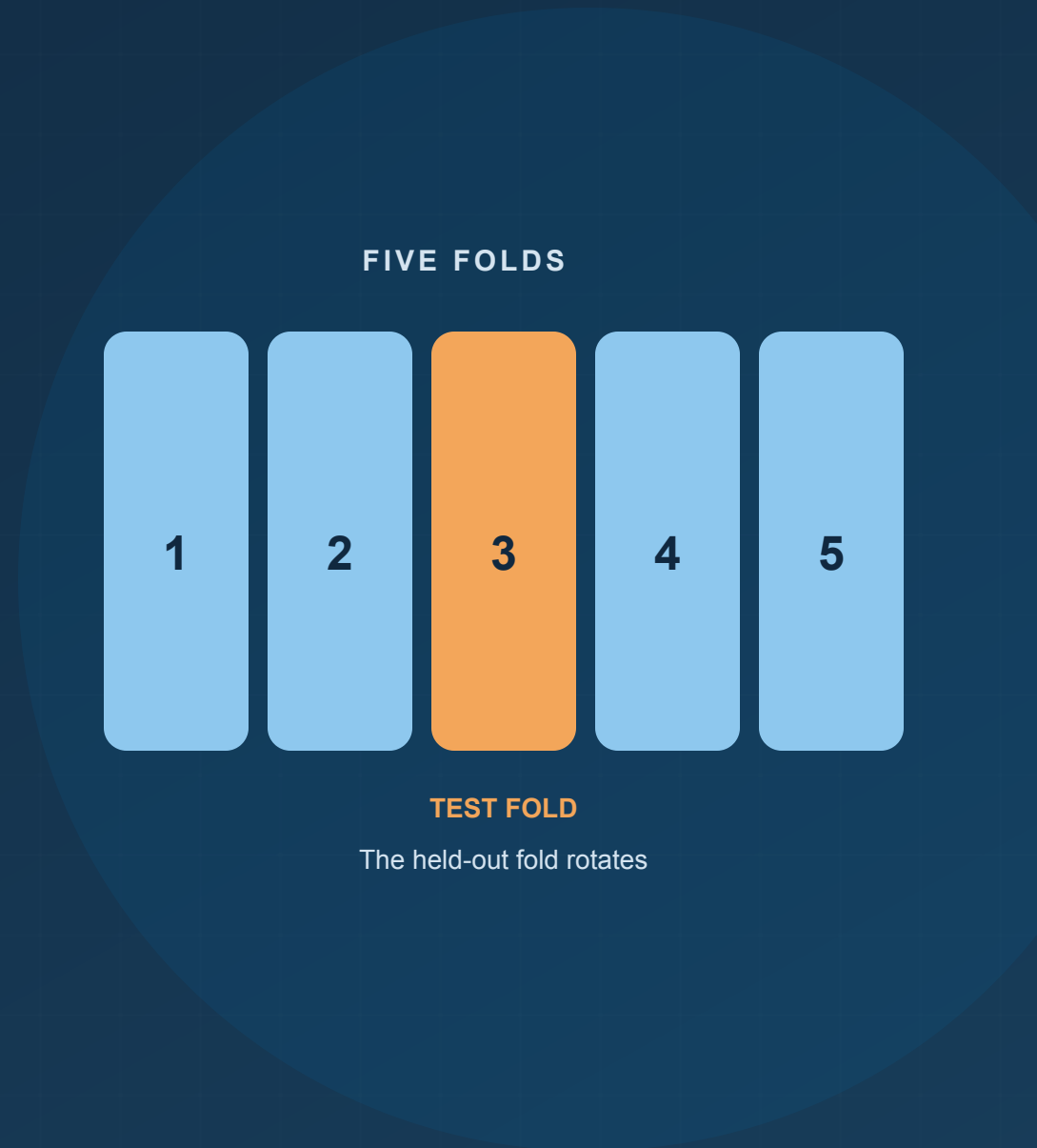


Same model. Same data. A different draw of the test rows.

PART 2

Cross-validation

Rotate the test fold so every row is tested once.



Split, rotate, average

1

Partition

Usually $k = 5$ or $k = 10$ equal folds.

2

Hold out one

Train on the other $k - 1$ folds.

3

Rotate

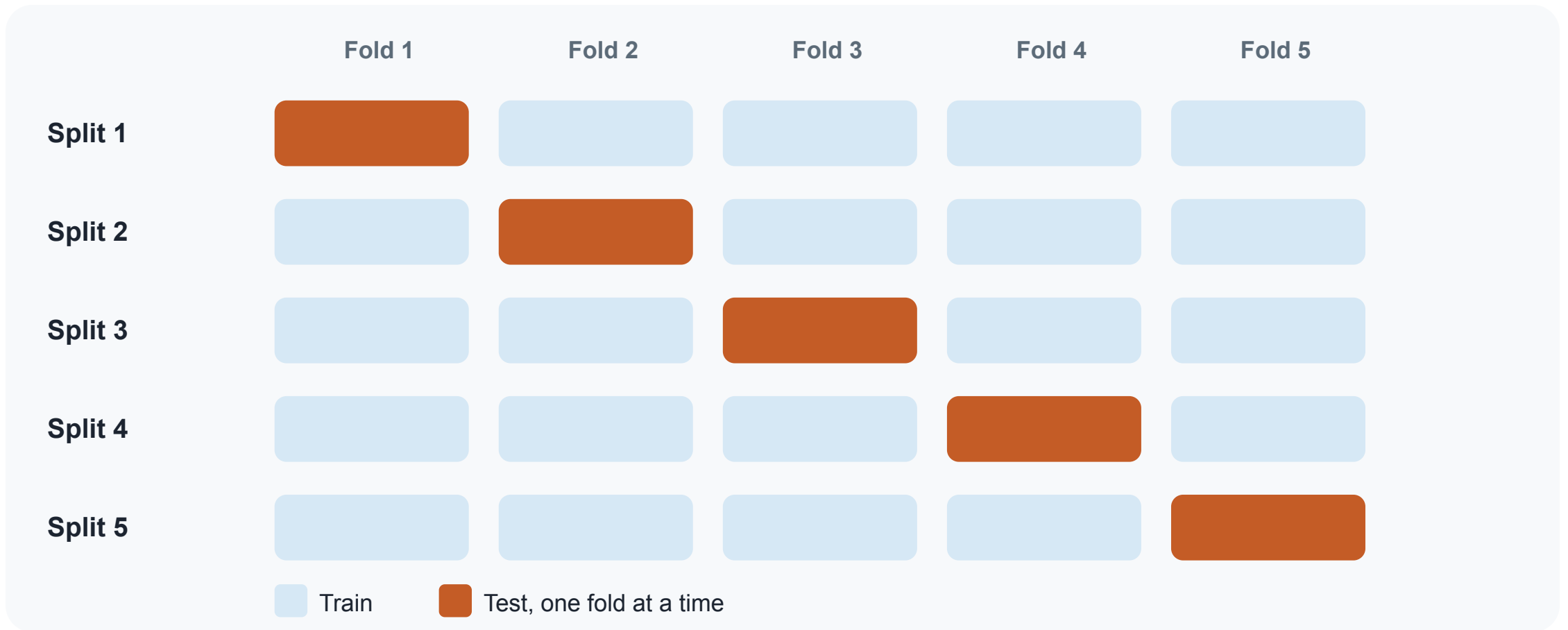
Each fold is the test set once.

4

Average

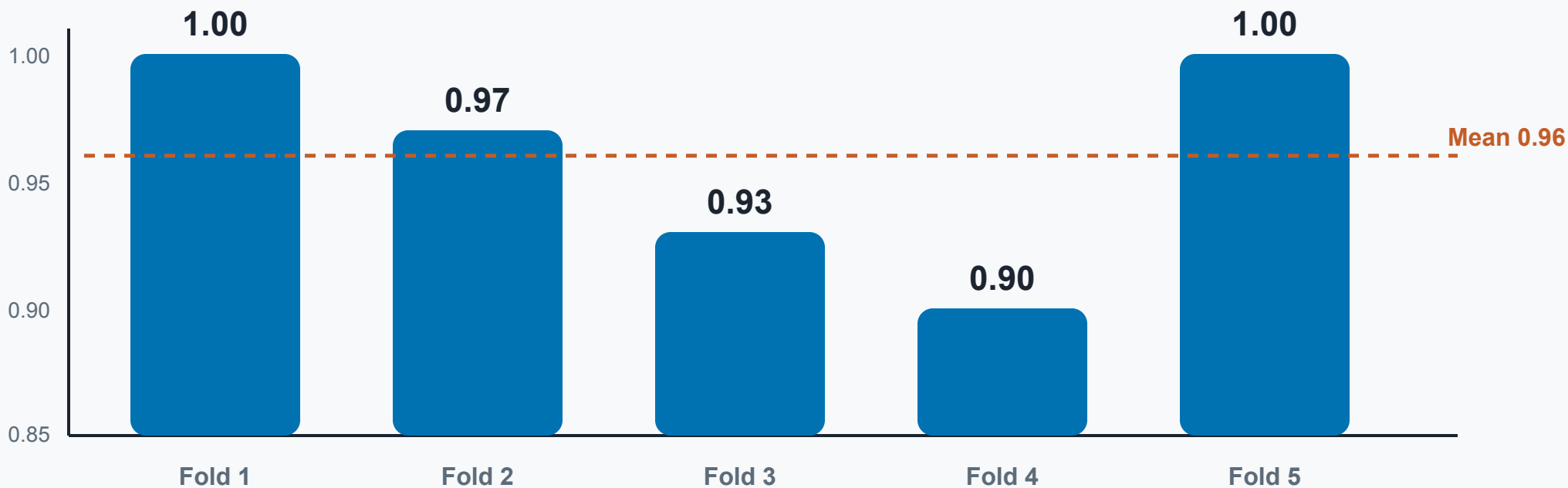
One score from the k test scores.

Five folds, five tests



A single split hides the range

what is y



One train/test split would have reported only one of these five numbers.

Scale starts at 0.85 so the gaps are visible. Iris, logistic regression.

Keep the class mix

PLAIN FOLDS, ROWS SORTED BY CLASS

Setosa, the whole block

Versicolor

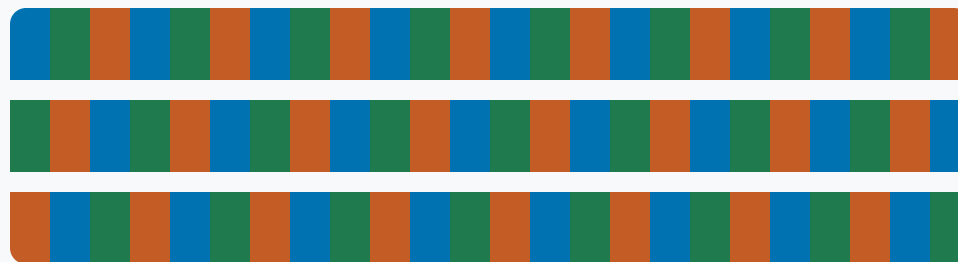
Virginica

0%

All three folds score 0%.

Each test fold is one whole class,
a class the model never saw.

STRATIFIED FOLDS



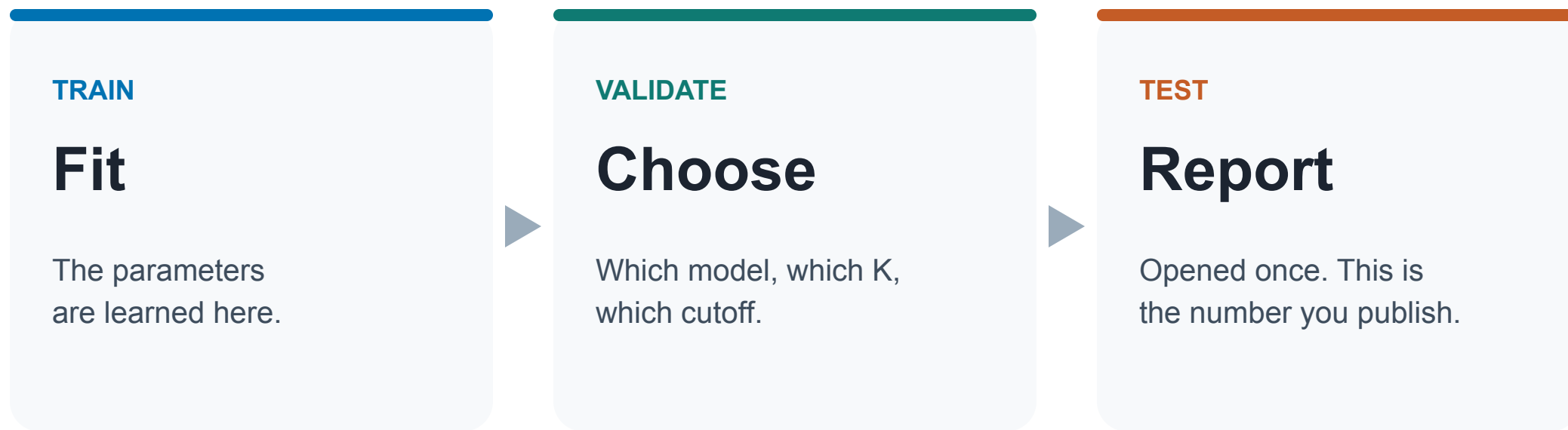
96%

Every fold keeps all three classes.

Same mix as the full data.

This is the default for classification.

Three jobs for the data



Validation may be one holdout, or the rotating folds of cross-validation.

PART 3

The confusion matrix

Every prediction is one of four outcomes.

1,000 loan applicants

		Predicted	
		No default	Default
Actual	No	900 True negative	50 False positive
	Yes	20 False negative	30 True positive

1,000 loan applicants. Positive class: will default.

What those four numbers say

		Predicted	
		No default	Default
Actual	No	900 True negative	50 False positive
	Yes	20 False negative	30 True positive

950 did not default

900 called correctly. 50 false alarms.

50 did default

30 were caught. 20 were missed.

80 were flagged

30 of those flags were real defaults.

A false alarm and a miss



FALSE POSITIVE

Type I error

A false alarm

A healthy patient is flagged.

Extra tests, extra stress.

Costly, and survivable.

In the loan table, this is the 50.



FALSE NEGATIVE

Type II error

A miss

A sick patient is cleared.

The disease goes untreated.

The error screening tries to avoid.

In the loan table, this is the 20.

PART 4

Beyond accuracy

Each metric answers a different question.

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{930}{1000} = 0.93$$

Useful

Balanced classes

And when a false alarm and a miss cost about the same.

Misleading

95% never default

Always saying no is already 95% accurate, and it catches nobody.

		Predicted	
		No default	Default
Actual	No	900 True negative	50 False positive
	Yes	20 False negative	30 True positive

Green cells are correct. The blue frame is everyone.
930 correct out of 1,000.

Precision

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{30}{80} = 0.375$$

37.5%

Of the flags were real

Fifty good customers are flagged anyway.

Use it when

A false alarm is expensive

A drug trial should start only when the signal looks real.

		Predicted	
		No default	Default
Actual	No	900 True negative	50 False positive
	Yes	20 False negative	30 True positive

Only the predicted-default column counts.
30 real defaults out of 80 flags.

Recall

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{30}{50} = 0.60$$

Higher precision

Fewer false alarms

More real defaults slip through.

Higher recall

Catch more defaults

More good customers are inconvenienced.

		Predicted	
		No default	Default
Actual	No	900 True negative	50 False positive
	Yes	20 False negative	30 True positive

Only the actual-default row counts.
We caught 30 of the 50 defaults.

Specificity

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{900}{950} \approx 0.947$$

0.947

Cleared the people who paid

Few good customers are blocked.

1 — Specificity

The false positive rate

That rate is the horizontal axis of the ROC curve.

		Predicted	
		No default	Default
Actual	No	900 True negative	50 False positive
	Yes	20 False negative	30 True positive

Only the actual no-default row counts.
We cleared 900 of the 950 who paid.

The F1 score

$$F_1 = \frac{2PR}{P + R}. \text{ With } P = 0.375 \text{ and } R = 0.60, F_1 \approx 0.462.$$

Harmonic mean

Both have to be decent

A model cannot earn a high F_1 by excelling at only one.

Left out

True negatives

One number still hides which side of the tradeoff you chose.

		Predicted	
		No default	Default
Actual	No	900 Not used	50 False positive
	Yes	20 False negative	30 True positive

Built from precision and recall.
The 900 true negatives are left out.

PART 5

The threshold

A probability becomes a class only after you pick a cutoff.

0.72 AGAINST 0.50

A score is not yet a label

Predicted $P(\text{default}) = 0.72$



Predict no default

Predict default

0.72 is above 0.50, so the label is Default.

Recall and precision trade places

Lower the cutoff to 0.30



6 of 8 flagged

Recall rises. More false alarms.

Raise the cutoff to 0.70



2 of 8 flagged

Precision rises. More misses.

● Predict no default ● Predict default

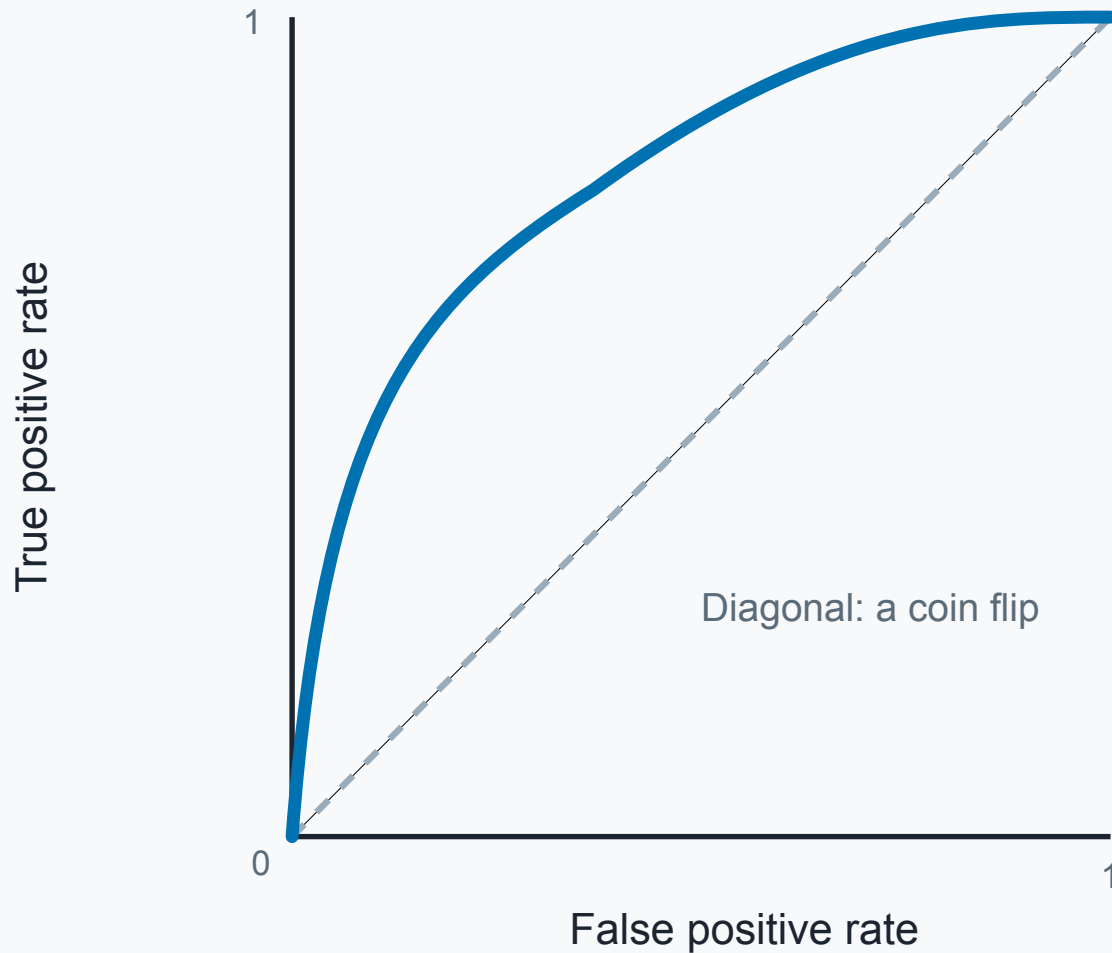
Same eight applicants. Only the line moved.

PART 6

ROC and AUC

Look at every cutoff, not just one.

True positive rate against false positive rate



Better: up and to the left

The curve stays above the diagonal.

Area under the ROC curve

1.0

Perfect ranking

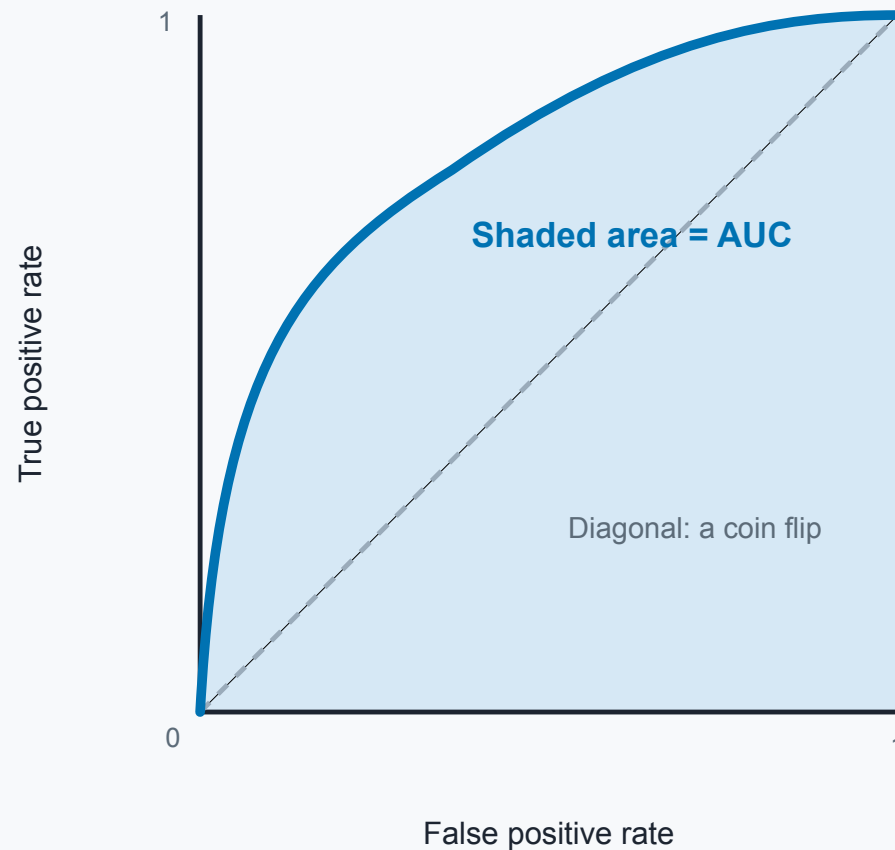
Every defaulter scores above every non-defaulter.

0.5

A coin flip

The model does not separate the classes better than chance.

AUC judges the ranking. It does not choose the cutoff you will use.



One cutoff, and all of them

At the cutoff you deploy

The decisions

Confusion matrix, accuracy, precision, recall, specificity, and F_1 .

Across cutoffs

The ranking

The ROC curve, and AUC as its summary.

From data to a number you can defend

1 Split

Hold out the test rows.

2 Fit

Train on the training set.

3 Tune

Validation or cross-validation.

4 Predict

Labels and probabilities.

5 Evaluate

Matrix, metrics, ROC.

6 Report

Name the cutoff and the metric.

What the code is scoring

```
y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[:, 1]
```

```
confusion_matrix(y_test, y_pred)
precision_score(y_test, y_pred)
recall_score(y_test, y_pred)
f1_score(y_test, y_pred)
roc_auc_score(y_test, y_prob)
```

y_pred

After the cutoff

Accuracy, precision, recall, and F_1 .

y_prob

Before the cutoff

ROC and AUC use the probabilities.

Six sentences

- 1 Grade the model on rows it did not use for fitting.
- 2 The confusion matrix separates hits, misses, and false alarms.
- 3 Accuracy misleads when one class dominates.
- 4 Precision, recall, specificity, and F_1 answer different questions.
- 5 The threshold trades false alarms against misses.
- 6 ROC shows every threshold. AUC summarizes the ranking.

READINGS

Chapter 5, and a market to practice on

ISLP, Chapter 5. Müller and Guido, Chapter 5.

Dataset: Smarket.csv. Direction is Up or Down. Next is the evaluation metrics lab.